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| Problem Chosen: | C             |

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### 2025 APMCM summary sheet

Regarding 2025 US "Reciprocal Tariffs" policy that significantly raises average tariffs on major trading partners and reshapes global trade cost structure, this paper constructs multi-layer quantitative assessment frameworks across soybeans, automobiles, semiconductors, tariff revenue, and macroeconomic manufacturing reshoring to evaluate impacts on US domestic industry .

For Problem 1, given China's differentiated preferences for soybean sources, we construct Armington demand model to characterize China's soybean import structure by source, combined with structural gravity and supply response models to describe global trade and production adjustments. Results show: as tariffs on US soybeans increase from 3% to 13%-25%, US export volume and value to China drop from 50% to 16%-26%, while Brazil and Argentina substantially increase market share through expansion and price adjustment, largely filling China's import gap.

For Problem 2, considering Japanese automakers' multi-location production (Japan, Mexico, US) and tariff response capabilities, we establish multi-production-path cost-minimization and nested Logit demand models linked with structural gravity and input-output frameworks to characterize trade and industry transmission. Results show: increased tariffs on Japanese vehicles shift production from Japan toward Mexico and US, boosting US and North American auto value-added and employment to some degree, yet average vehicle prices rise two percentage points, and import share declines noticeably.

For Problem 3, we construct multi-stage global semiconductor supply chain network flow model incorporating downstream demand and dynamic capacity evolution equations, simulating chip manufacturing layout and trade flows under various tariff and export control combinations. Results show: high tariffs partially replacing subsidies significantly raise US mid-high-end process capacity and domestic manufacturing share, yet require higher fiscal and downstream costs; global supply chain concentration rises with increased dependence on allied regions, reflecting clear efficiency-security tradeoffs.

For Problem 4, from HS item level we construct partial-equilibrium tariff revenue model with structural gravity trade-shift module and dynamic Laffer-effect adjustment model. Results show: reciprocal tariffs initially boost revenue substantially (exceeding 40% in some years), but with trade contraction and partner retaliation, cumulative four-year revenue gains are limited, even falling below baseline in some scenarios.

For Problem 5, we construct multi-sector dynamic input-output and trade shock model combined with manufacturing reshoring decisions and Manufacturing Reshoring Index (MRI), embedding simplified financial market shock module to transmit tariffs and countermeasures through US agriculture, manufacturing, services, and finance. Results show: high tariffs and countermeasures modestly raise US manufacturing capacity and MRI, yet increase prices, compress external demand, suppress investment; GDP and employment gains are limited and unsustainable; actual manufacturing "reshoring" magnitude is moderate, far from policy-claimed "large-scale reshoring."

**Keywords:** Armington Model   Structural Gravity Model   Nested Logit Model   Network Flow Model   Dynamic Input-Output Model

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# I. Problem Restatement

## 1.1 Problem Background

April 2025: US government announces "Reciprocal Tariffs" policy imposing minimum 10% baseline tariffs on all trading partners, with additional 11%-50% tariffs on 60 countries with large US trade deficits. Under this backdrop, global trade patterns and supply chains face reshaping: agricultural trade (e.g., soybeans) shows clear shifts; Japanese automakers pursue tariff avoidance through "localization + third-country layout"; semiconductor industry faces heightened supply chain security and tech competition risks amid combined tariffs, subsidies and export controls; US tariff revenue surges short-term, but long-term impacts of import cost increases, trade contraction and alliance shifts remain unclear.

## 1.2 Problem Requirements

**Problem 1: Soybean trade patterns and tariff impacts.** Model soybean export volumes, values and market share redistribution among US, China, Brazil and Argentina following US reciprocal tariffs and countermeasures.

**Problem 2: US-Japan auto trade and non-tariff responses.** Analyze Japan's market position, non-tariff responses and economic transmission mechanisms; assess changes in US-Japan auto trade, US import structure, and US domestic auto industry from increased Japanese car tariffs.

**Problem 3: Semiconductor industry security and tariff policy.** Model different chip grades' trade and production amid combined tariffs, subsidies and high-end export controls; analyze tariff impacts on US domestic manufacturing capacity, industry profits and supply chain security.

**Problem 4: Dynamic tariff revenue changes and second-term forecasts.** Build HS-level dynamic model of tariff revenue and trade volume; assess tariff revenue paths and ranges during Trump's second term, considering trade elasticity and policy uncertainty.

**Problem 5: Macroeconomic comprehensive effects and manufacturing reshoring assessment.** Build comprehensive model capturing sectoral output, employment, trade balances and financial market fluctuations; assess whether reciprocal tariffs actually promote US manufacturing reshoring short-to-medium term and whether macro costs are acceptable.

# II. Problem Analysis

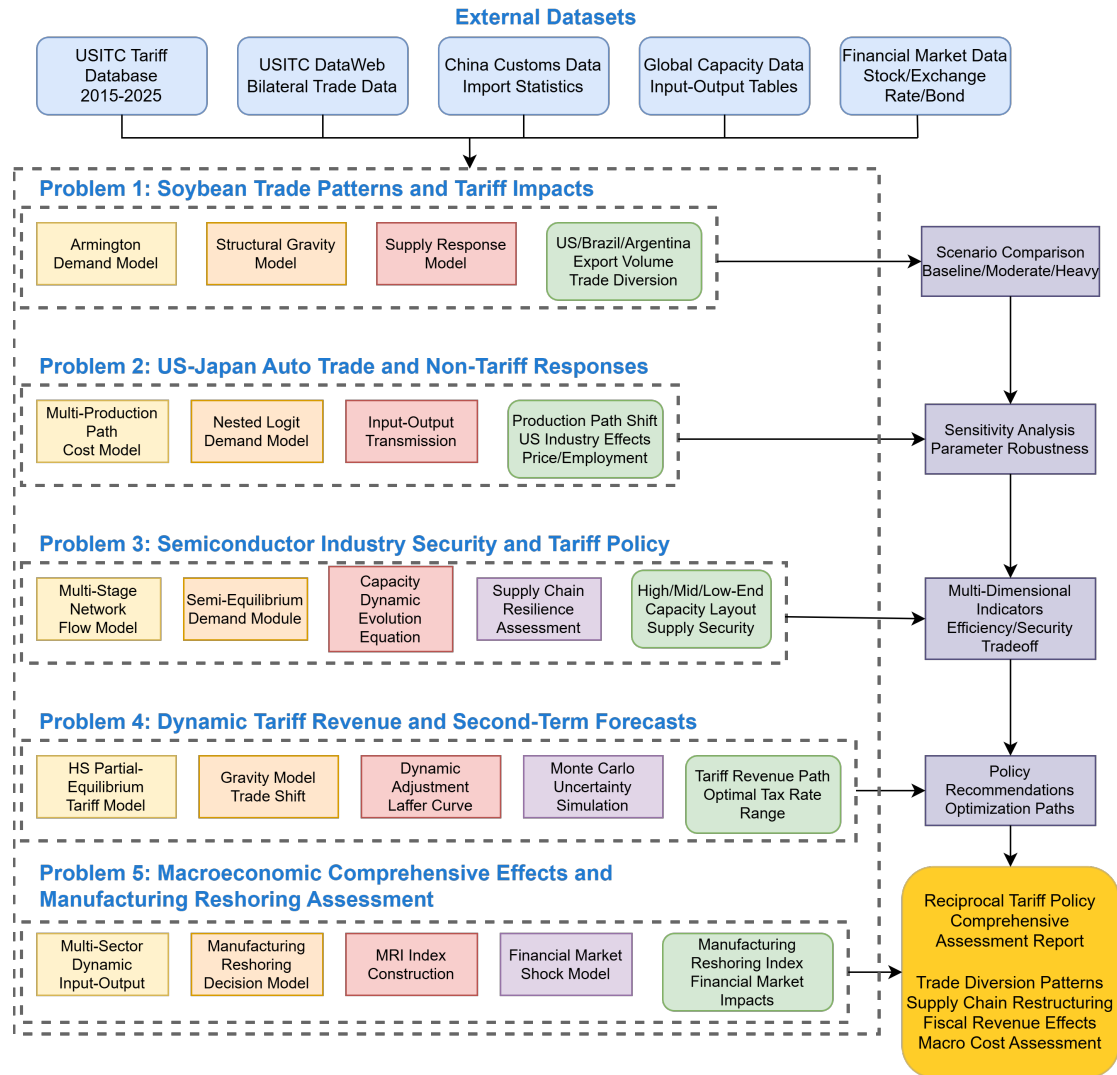
## 2.1 Problem 1 Analysis

From demand-side using Armington import structure model for China's soybean consumption shares from different sources; from supply-side using structural gravity and simplified supply response models to estimate tariff-scenario trade redistribution among US, Brazil, Argentina.

## 2.2 Problem 2 Analysis

Build brand-location dual market share model and cost-price transmission model including tariffs, shipping, exchange rates and non-tariff costs, combined with input-output analysis, charac-





**Figure 1 Overall Methodology Flowchart**

terizing Japanese automakers' non-tariff responses (localization, product mix, pricing) impacts on US-Japan trade, US import structure, and domestic auto industry.

### 2.3 Problem 3 Analysis

Build multi-stage global semiconductor supply chain network flow model characterizing capacity and trade reconfiguration across economies and process nodes under cost minimization and risk constraints; describe high-end manufacturing capability development through technology transition and capacity evolution equations; construct supply chain resilience metrics evaluating industry security under extreme scenarios.

### 2.4 Problem 4 Analysis

Build partial-equilibrium tariff revenue model at HS item level; estimate tariff elasticities of bilateral trade using structural gravity; introduce gradual adjustment and stochastic shocks simulating Trump's second-term tariff revenue dynamics; apply simplified Laffer curve framework character-

izing nonlinear rate-revenue relationships.

## 2.5 Problem 5 Analysis

Core multi-sector dynamic input-output model with manufacturing reshoring decision model and Manufacturing Reshoring Index (MRI), characterizing US capacity share changes in global and high-tech manufacturing; analyze tariff and countermeasure impacts on dollar, Treasury yields and risk asset prices through simplified rate-exchange-asset price module, providing unified comprehensive effect assessment.

## III. Model Assumptions

Without affecting main conclusions, this paper makes following overall assumptions:

- **Assumption 1:** Tariff rates, trade values and prices unified in US dollars; short-term nominal exchange rate fluctuations primarily absorbed through prices.
- **Assumption 2:** Each country's agriculture, auto sectors have stable technology and capacity; only semiconductor high-end capacity ramp explicitly modeled; other sectors' production functions given.
- **Assumption 3:** Problems 1-3 use partial equilibrium at industry level with representative producers; macro and other sectors reflected indirectly. Problem 5 adds multi-sector input-output approximating macro structure.
- **Assumption 4:** Tariffs and countermeasures primarily affect trade flows and capacity through CIF prices and firm cost-profit changes; other non-tariff measures converted to equivalent tariffs or cost markups; policy expectations reflected through scenarios and sensitivity analysis.

## IV. Notation

| Symbol                  | Definition                                                          |
|-------------------------|---------------------------------------------------------------------|
| $i, j$                  | Country/region index (e.g., US, CN, BR, AR)                         |
| $g$                     | Commodity group or sector index (e.g., soybean, auto, chip)         |
| $\tau_{ij,t}$           | Period $t$ composite ad-valorem tariff from $i$ to $j$              |
| $M_{ij,t}$              | Period $t$ import value from $i$ to $j$                             |
| $Q_{ij,t}$              | Period $t$ import quantity from $i$ to $j$                          |
| $P_i^{\text{FOB}}$      | Country $i$ export FOB price                                        |
| $C_i$                   | China's comprehensive CIF cost for importing soybean from $i$       |
| $s_i$                   | Armington import share from $i$ to China                            |
| $\varepsilon_d, \sigma$ | Demand price elasticity and source substitution elasticity          |
| $X_{g,i,t}$             | Period $t$ export volume or value of commodity $g$ from country $i$ |
| $R_t$                   | Period $t$ total US tariff revenue                                  |
| $\mathbf{x}_t$          | Output vector by sector in period $t$                               |
| $A$                     | Direct consumption coefficient matrix                               |

## V. Problem 1: Model Development and Solution

### 5.1 Data Preparation and Processing

Main data sources:

- **Tariff Data (US perspective):** USITC Tariff Database 2015-2025, extracting soybean HS codes (12011000, 12019000) MFN rates, preferential rates and collection methods, providing MFN\_AD\_VAL\_RATE, MFN\_RATE\_TYPE\_CODE fields for equivalent ad-valorem rate (AVE) calculation.
- **US Bilateral Trade Data:** USITC DataWeb exports for US-China, Brazil, Argentina soybean trade (FAS Value) and import tariff collections, 2015-2024, cross-checked with UN Comtrade.
- **China's Imports and Global Supply:** China Customs data on soybean imports by source, FAOSTAT data on US, Brazil, Argentina soybean output and exports, CBOT futures prices as world reference.

Scope: soybean raw grain under HS1201, excluding processed soybean oil and meal. Raw data unified: unit standardization, missing value handling, mirror adjustment, baseline data construction.

USITC Tariff Database contains tax types: duty-free, pure ad-valorem, pure specific, and mixed types. For model uniformity, convert all to equivalent ad-valorem rate (AVE):

- When MFN\_RATE\_TYPE\_CODE = 0: MFN rate is Free, AVE = 0;
- When MFN\_RATE\_TYPE\_CODE = 7: Pure ad-valorem, AVE = MFN\_AD\_VAL\_RATE;
- Other mixed types: convert specific portion to "percentage of import value," add to ad-valorem portion for total AVE.

For HS1201 soybeans, US MFN rate historically 0

$$\tau_{ij} = \tau_{ij}^{\text{MFN}} + \tau_{ij}^{\text{extra}}.$$

### 5.2 Multi-Model Framework

Three complementary perspectives: **China's import structure view, global bilateral trade view, industry supply response view.**

#### 5.2.1 Model 1: Armington-Based China Import Structure Model

Assume different-country soybeans in China are "substitutable but imperfect" Armington goods with two-layer structure:

- **China's Total Soybean Import Demand:**

$$M_{CN} = A_{CN} (P_{CN}^{\text{agg}})^{-\varepsilon_d}, \quad (1)$$

where  $M_{CN}$  = China's total soybean imports,  $P_{CN}^{\text{agg}}$  = source-weighted average CIF price,  $\varepsilon_d$  = demand price elasticity,  $A_{CN}$  = scale parameter.

- **Armington Share Structure:**

$$s_i = \frac{\exp(-\sigma C_i + \theta_i)}{\sum_k \exp(-\sigma C_k + \theta_k)}, \quad (2)$$

where  $C_i$  = source country  $i$  comprehensive import cost,  $\sigma$  = price sensitivity (substitution elasticity),  $\theta_i$  = quality and preference fixed effect.

Comprehensive cost consists of FOB price, tariffs and freight:

$$C_i = P_i^{\text{FOB}}(1 + \tau_{CN,i}) + f_{CN,i}, \quad (3)$$

where  $f_{CN,i}$  = average transport and non-tariff costs.

Given tariff scenarios, China's imports from  $i$  are

$$Q_{i \rightarrow CN} = s_i \cdot M_{CN},$$

with export value  $V_{i \rightarrow CN} = Q_{i \rightarrow CN} \cdot P_i^{\text{FOB}}$ .

### 5.2.2 Model 2: Structural Gravity Model

Standard log-linear structural gravity model for bilateral soybean trade:

$$\ln X_{ij,t} = \alpha + \beta \ln(1 + \tau_{ij,t}) + \delta_{i,t} + \eta_{j,t} + \gamma^\top Z_{ij} + \epsilon_{ij,t}, \quad (4)$$

where  $Z_{ij}$  = gravity variables (distance, borders, language),  $\delta_{i,t}$ ,  $\eta_{j,t}$  = exporter and importer time fixed effects. Parameter  $\beta < 0$  represents tariff suppression; agriculture literature suggests  $\hat{\beta} \approx -3.5$  as baseline.

Under reciprocal tariffs scenario, tariff changes from  $\tau_{ij}^{(0)}$  to  $\tau_{ij}^{(k)}$  give:

$$X_{ij}^{(k)} = X_{ij}^{(0)} \cdot \exp \left[ \hat{\beta} \left( \ln(1 + \tau_{ij}^{(k)}) - \ln(1 + \tau_{ij}^{(0)}) \right) \right]. \quad (5)$$

### 5.2.3 Model 3: Supply Response and Industry Impact Model

Simplified supply response model for export countries  $i$ :

$$S_i(P_i) = S_i^0 \left( \frac{P_i}{P_i^0} \right)^{\eta_i}, \quad (6)$$

where  $\eta_i$  = supply price elasticity (short-medium term: 0.2-0.4). Tariff-induced world price and demand changes allow estimating output and export income adjustments.

## 5.3 Policy Scenario Construction

Using 2024 baseline, we define:

$$s_i^0 = \frac{Q_{i \rightarrow CN}^0}{\sum_k Q_{k \rightarrow CN}^0}, \quad C_i^0 = \frac{V_{i \rightarrow CN}^0}{Q_{i \rightarrow CN}^0}.$$

After solving for Armington preference parameters  $\theta_i$  and scale parameter  $A_{CN}$ , we construct three scenarios:

- **Scenario 0 (Baseline)**: Current tariff structure; China's tariff on US soybeans 3%, Brazil/Argentina 0%;
- **Scenario 1 (Moderate)**: US reciprocal tariffs trigger moderate China response; China's tariff on US soybeans 13%, Brazil/Argentina 0%;

- **Scenario 2 (Heavy):** Trade tensions escalate; China's tariff on US 25%, Brazil/Argentina reduced by 2% (e.g., -2%) to strengthen alternatives.

For each scenario  $k$ , update tariff vector  $\tau^{(k)}$ , recalculate  $C_i^{(k)}$ ,  $s_i^{(k)}$ ,  $M_{CN}^{(k)}$ , obtaining  $Q_{i \rightarrow CN}^{(k)}$ ,  $V_{i \rightarrow CN}^{(k)}$ .

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**Algorithm 1** Reciprocal Tariffs Scenario: US-China Soybean Trade
 

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**Input:** Baseline trade data  $\{Q_{i \rightarrow CN}^0, V_{i \rightarrow CN}^0\}$ , FOB prices  $P^{\text{FOB}}$ , shipping costs  $f$ , tariff scenarios  $\{\tau^{(k)}\}$ , parameters  $\sigma, \varepsilon_d, A_{CN}, \beta$

**Output:** Scenario shares  $s_i^{(k)}$ , imports  $Q_{i \rightarrow CN}^{(k)}$ , exports  $V_{i \rightarrow CN}^{(k)}$ , tariff revenue and concentration metrics

- 1: Back out  $\theta_i$  from baseline data using Eq. (2);
  - 2: **for** each scenario  $k$  **do**
  - 3:   Update tariff vector  $\tau^{(k)}$ , compute comprehensive costs  $C_i^{(k)}$ ;
  - 4:   Call Armington model: compute  $s_i^{(k)}$ ,  $M_{CN}^{(k)}$ ,  $Q_{i \rightarrow CN}^{(k)}$ ,  $V_{i \rightarrow CN}^{(k)}$ ;
  - 5:   Calculate China's tariff revenue  $R^{(k)} = \sum_i \tau_{CN,i}^{(k)} \cdot P_i^{\text{FOB}} \cdot Q_{i \rightarrow CN}^{(k)}$ ;
  - 6:   Calculate import concentration HHI:  $HHI^{(k)} = \sum_i (s_i^{(k)})^2$ ;
  - 7:   Use gravity model Eq. (5) to forecast  $X_{US,CN}^{(k)}$ , compare with  $V_{US \rightarrow CN}^{(k)}$  for robustness check;
  - 8:   Use supply model Eq. (6) to estimate three countries' supply and revenue changes.
  - 9: **end for**
- 

## 5.4 Multi-Dimensional Results and Analysis

### (1) Armington Model: US-China Soybean Trade Restructuring

Armington model (Figure 2) shows: as China's tariffs escalate from moderate to heavy retaliation, US soybean exports and market share drop significantly (export quantity cumulative decline 70%, share from 50% to 16%); Brazil and Argentina exports rise simultaneously, with Brazil as main beneficiary and Argentina providing supplemental gains. China's total soybean import volume remains stable, but import structure shifts from "US-centered" to "Brazil-centered with Argentina as supplement," showing typical tariff-driven trade diversion rather than contraction.

### (2) Structural Gravity Model: Alternative Tariff Elasticity Characterization

Gravity model (Figure 3) shows: under same tariff path, US soybean exports to China in strong and heavy retaliation scenarios both drop 60%-70%, while Brazil and Argentina remain relatively stable with slight increases, showing strong "sanctioned country" shock—in extreme scenarios US export quantity declines 70%, Brazil and Argentina changes controlled within 10%. Trade elasticity illustration shows tariff elasticity  $\beta = -4$  yields 70% US export decline if tariff increases from 0% to 40%.

### (3) Supply-Demand Equilibrium Model: Global Price and Quantity Effects

Supply-demand equilibrium model (Figure 4) shows: even under extreme tariff scenarios, total US, Brazil, Argentina soybean exports decline modestly; output index remains above 2020 levels, indicating long-term global demand dominates with tariffs causing regional trade redistribution rather than global contraction. As China shifts purchases from US to South America, world soybean

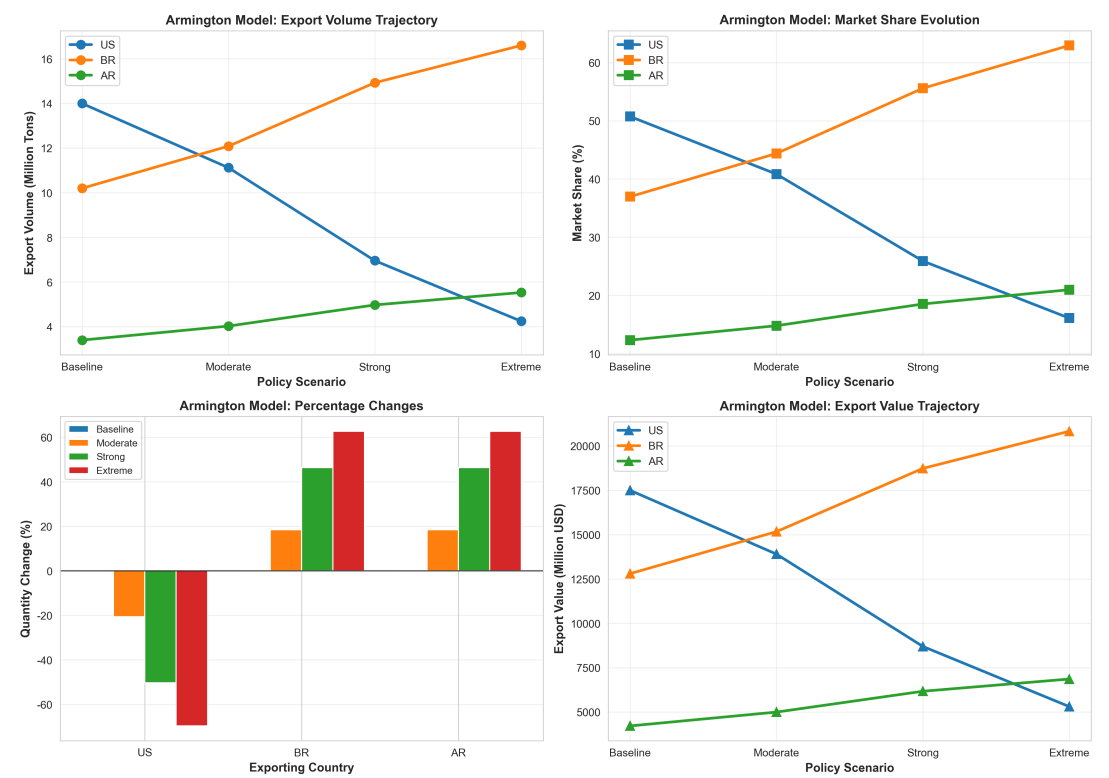


Figure 2 Armington Model: Soybean Export Volume, Market Share and Value Evolution

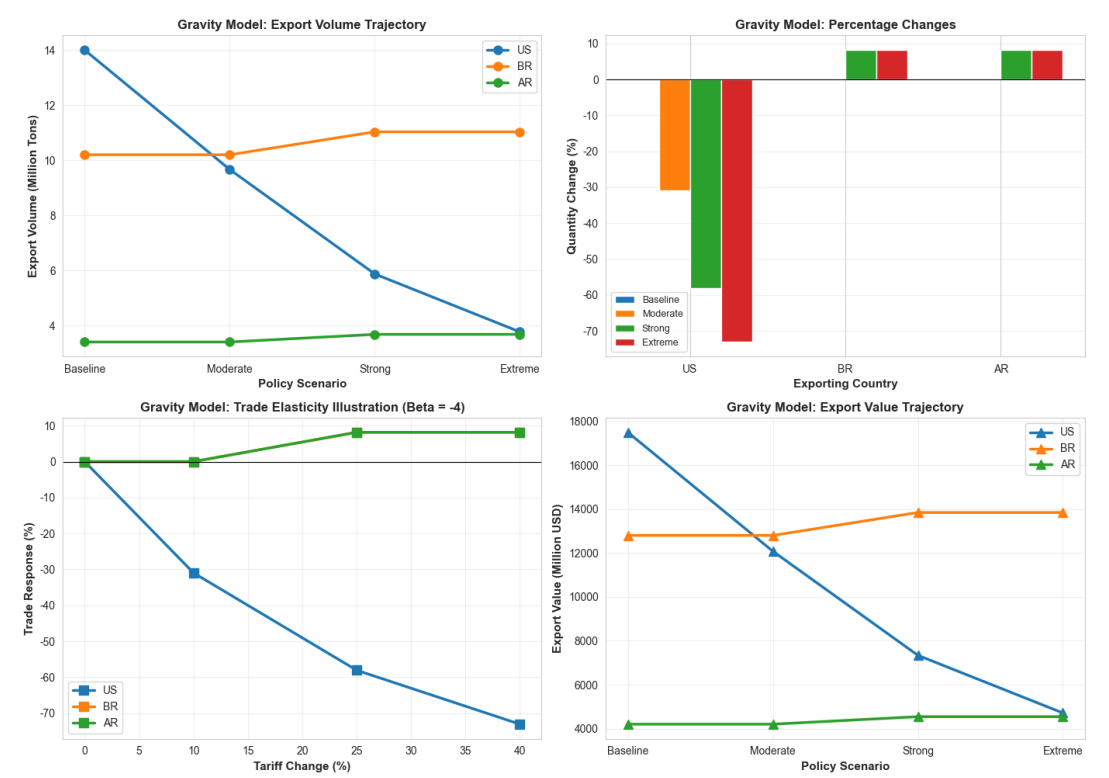
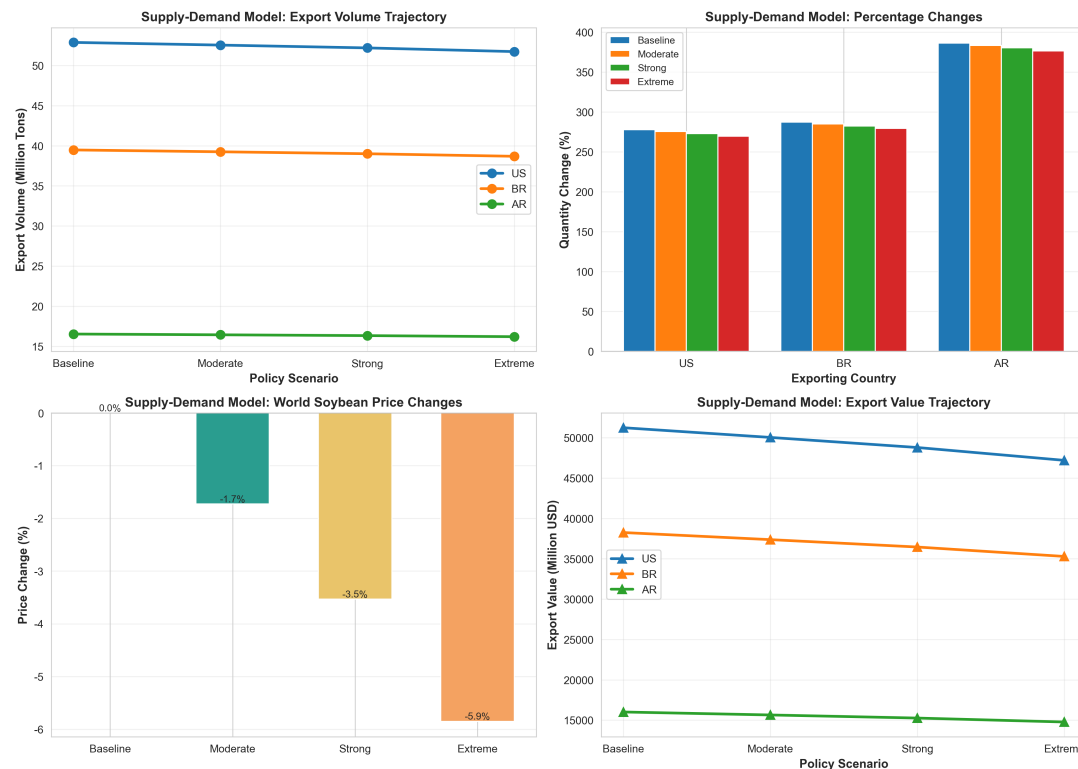


Figure 3 Structural Gravity Model: Export Volume, Value and Tariff Elasticity Analysis



**Figure 4 Supply-Demand Equilibrium: Global Export Volume, Price and Value Changes**

reference price declines 1.7%, 3.5%, 5.9% under moderate, strong and extreme scenarios respectively, squeezing three countries' export prices. US faces "quantity and price decline" double losses in market share and revenue, while Brazil and Argentina benefit from "quantity increase, price decline" with continued modest revenue growth.

#### ***(4) Multi-Model Comprehensive Comparison and Policy Implications***

Figure 5 compares three models' results across six panels: as US-China tariff gap widens from 2 to 40 percentage points, US soybean exports drop from 14 MMT to 4 MMT, market share declines from 50% to 16%, export value cumulative decline nearly 75%; correspondingly, Brazil and Argentina exports reach 16.7 MMT and 5.6 MMT, with 5%-10% positive export growth and combined share exceeding 80%. All three models agree in strong and extreme scenarios: US export reductions 50% or higher, Brazil/Argentina gains in 40%-60% range; Armington heatmap highlights "higher tariffs drive nonlinear US losses, South American gains approach their losses" pattern. Overall, sustained large tariff gaps will stably shift China's structure from US toward Brazil-led South America; South American gains basically offset US losses.

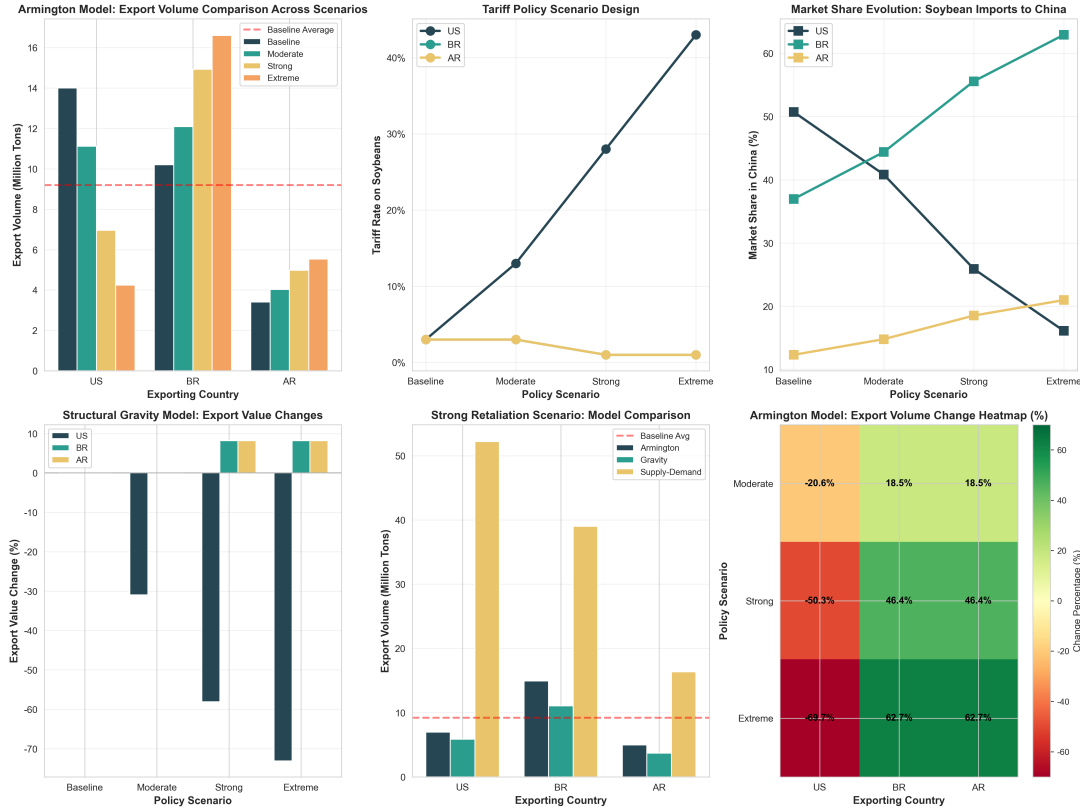


Figure 5 Soybean Trade Multi-Model Comprehensive Results and Tariff Scenario Comparison

## VI. Problem 2: Model Development and Solution

### 6.1 Data Preparation and Current Status Characterization

We define "automobiles" as passenger vehicles and light trucks (HS 8703.\*), optionally including select light commercial vehicles (8704.\*). Based on USITC DataWeb and industry reports, we construct 2024 US auto market baseline data:

- **Overall Scale:** 2024 US light vehicle sales  $Q_{\text{tot}}^{\text{US}}$  units, 46% imported, rest domestic production;
- **Import Structure:** By source country/economy (Japan, Mexico, EU, Korea, others) with import value and quantity;
- **Brand and Location Structure:** Japanese brands in US further split into three categories —
  1. Japan domestic production → direct US export;
  2. Mexico low-cost production → US export (USMCA preference);
  3. US domestic factories (localization).

Baseline calibration uses Japanese brands' US sales volume  $Q_{\text{US}}^{\text{JP}}$  with path shares:

$$s_{\text{JP} \rightarrow \text{US}}^0 = 43.6\%, \quad s_{\text{MX} \rightarrow \text{US}}^0 = 38.5\%, \quad s_{\text{US-local}}^0 = 17.9\%,$$

with Japanese brands' overall market share 13%, US-Japan auto trade value \$60.5B baseline (2024).



## 6.2 Multi-Model Framework

### 6.2.1 Model 1: Japanese Multi-Production-Path Cost Minimization

Japanese brands' US sales  $Q_{US}^{JP}$  supplied through three paths:

$$\mathcal{P} = \{JP \rightarrow US, MX \rightarrow US, US\text{-local}\}.$$

Unit comprehensive cost to US market for path  $p$ :

$$C_p = \begin{cases} (c_p + t_p)(1 + \tau_p), & p \in \{JP \rightarrow US, MX \rightarrow US\}, \\ c_{US}, & p = US\text{-local}, \end{cases} \quad (7)$$

where  $\tau_p$  = effective tariff rate for path  $p$ .

Firms flexibly allocate across paths using Armington/logit "soft minimization," with path sales shares:

$$s_p = \frac{\exp(-\gamma C_p + \psi_p)}{\sum_{q \in \mathcal{P}} \exp(-\gamma C_q + \psi_q)}, \quad (8)$$

where  $\gamma > 0$  = cost sensitivity,  $\psi_p$  = path preference (absorbs institutions, supply chain stickiness).

Given total sales  $Q_{US}^{JP}$ , path volumes are:

$$Q_p = s_p \cdot Q_{US}^{JP}. \quad (9)$$

To capture "Japanese offset tariffs via localization and Mexico capacity," we add soft constraints:

$$s_p \leq \bar{s}_p(t),$$

where  $\bar{s}_p(t)$  = time-evolving capacity limits. In medium-term scenarios,  $\bar{s}_{US\text{-local}}$  and  $\bar{s}_{MX \rightarrow US}$  increase modestly, reflecting Japanese investment expansion.

### 6.2.2 Model 2: US Auto Demand Price Elasticity

As tariffs raise Japanese car costs, sales respond to prices. Using constant-elasticity demand:

$$Q_{US}^{JP} = Q_0^{JP} \left( \frac{P^{JP}}{P_0^{JP}} \right)^{\varepsilon_{JP}}, \quad (10)$$

where  $\varepsilon_{JP} = -0.8$  = Japanese brand price elasticity in US market.

Japanese brand average price via cost-plus markup:

$$P^{JP} = (1 + \mu) \cdot \sum_{p \in \mathcal{P}} s_p C_p, \quad (11)$$

where  $\mu$  = profit markup. We introduce **price pass-through rate**  $\theta_{\text{pass}} \in [0, 1]$  showing cost increase share passed to final prices, remainder absorbed by profit compression:

$$P^{JP} = P_0^{JP} + \theta_{\text{pass}} \cdot \left( \sum_p s_p C_p - \sum_p s_p^0 C_p^0 \right).$$

Similarly construct US and EU brand elasticity relations; in quantitative implementation focus on Japanese brand changes, treating others as aggregate.

### 6.2.3 Model 3: US-Japan Bilateral Trade Gravity Model

From macro trade perspective, construct structural gravity model for Japan-US auto trade elasticity. For Japan's auto exports to US:

$$\ln X_{JP \rightarrow US} = \alpha + \beta \ln(1 + \tau_{JP \rightarrow US}) + \delta_{JP} + \eta_{US} + \gamma^T Z_{JP,US} + \varepsilon, \quad (12)$$

where  $\tau_{JP \rightarrow US}$  = effective tariff on Japanese vehicles,  $Z_{JP,US}$  = gravity variables (distance, agreements). Auto trade literature suggests tariff elasticity  $\beta = -2.5$  (1% tariff increase  $\rightarrow$  2.5% trade decline).

In scenario analysis, use approximation:

$$X_{JP \rightarrow US}^{(k)} = X_{JP \rightarrow US}^{(0)} \cdot \exp [\beta (\ln(1 + \tau^{(k)}) - \ln(1 + \tau^{(0)}))], \quad (13)$$

### 6.2.4 Model 4: US Auto Industry Input-Output Transmission

Simple input-output framework transmits "vehicle production changes" to upstream (parts, steel, metals, etc.). With sectors: vehicle manufacturing, auto parts, steel/metals, other manufacturing, services/consumption. Using input-output coefficient matrix  $A$ :

$$X = AX + Y \quad \Rightarrow \quad X = (I - A)^{-1}Y.$$

Tariff scenarios yield US domestic vehicle output change  $\Delta X_1$  and import substitution  $\Delta M$ , transforming to final demand change  $\Delta Y_1$ :

$$\Delta X = (I - A)^{-1} \Delta Y,$$

estimating parts, steel industry output and employment changes.

## 6.3 Policy Scenario Construction

Three typical scenarios:

- **Scenario 0 (Baseline):** Current Japan-US agreement tariffs; Japanese brand paths (43.6%, 38.5%, 17.9%); US-Japan trade value \$60.5B;
- **Scenario 1 (Moderate):** US adds 15% on Japanese passenger cars, 10% on Mexico cars, USMCA benefits retained;
- **Scenario 2 (Heavy):** 35% on Japan cars, 20% on Mexico, limiting "Mexico tariff bypass" space.

Each scenario allows Japanese firms to offset via US localization and Mexico capacity expansion (capacity limits  $\bar{s}_p$  increase), models automatically adjust actual path shares.

## 6.4 Multi-Dimensional Results Analysis

### (1) Japanese Firm Strategy Comparison: Cost, Price and Comprehensive Effects

Figure 6 under 35% tariffs compares Japanese firm response strategies' cost-benefit tradeoffs: Panel A shows Mexico transshipment and US localization significantly lower unit comprehensive cost versus Japan direct supply; Panel B shows localization rate 0% to 30% rapidly raises net returns

and ROI, optimal 30%; Panel C reveals price pass-through rate tradeoff between profit erosion and demand loss, total impact minimized 70% pass-through; Panel D shows considering market share, profit rate, demand shock and sustainability, "location restructuring + localization investment + modest pricing" optimal, followed by "localization + modest restructuring," pricing-only worst long-term.

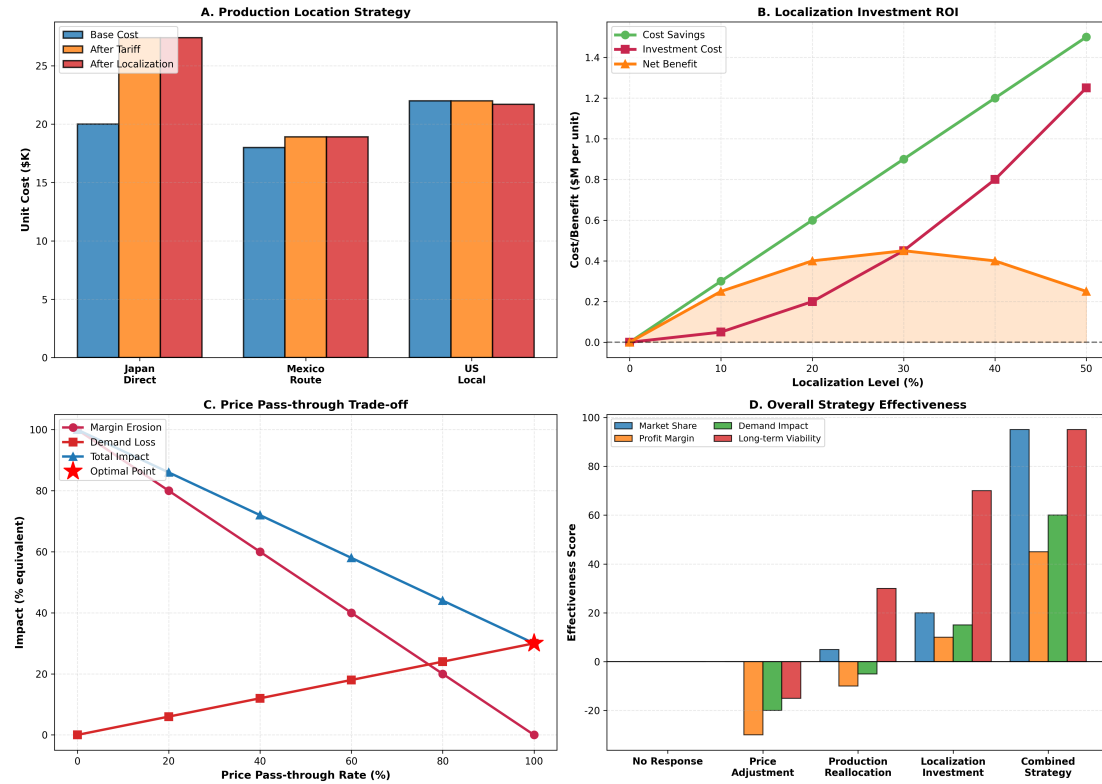


Figure 6 Japanese Auto Firm Strategy Effects Across Tariff Scenarios

## (2) US Market Competition: Brand Share Stability and Fine-Tuning

Figure 7 heatmap shows across baseline (2%), moderate (+15%), heavy (+35%) tariff scenarios, US auto market brand structure nearly unaffected: Japanese brand share stable 14%, US brands 67%, EU/Korea/others vary only decimal places. Even 35% Japan tariff causes only  $\pm 0.02$  percentage point share change, below statistical error. Result shows: with "location restructuring + localization" mechanisms functioning, tariff costs highly internalized and offset; consumer-level prices and choices nearly unaffected; brand competition structure highly stable short-term.

## (3) Elasticity and Parameter Sensitivity: Model Robustness Tests

Figures 8 and 9 show model demonstrates good robustness within reasonable parameter ranges: tariff 0-50% with elasticity  $\varepsilon \in [-2.5, -1.2]$ , Japan direct imports decline 40%-65% in monotonic pattern; localization rate responds approximately linearly to tariff changes. Within price elasticity -0.8 to -2.0, Japanese brand share sensitivity only 0.01 percentage point magnitude, reflecting strong brand loyalty buffer. Tornado diagram shows tariff elasticity and FDI adjustment speed most affect Japan direct imports ( $\pm 2\%$ -3%), price elasticity and costs next, brand loyalty weakest, indicating conclusions mainly sensitive to tariff elasticity but robust to brand preference assumptions.

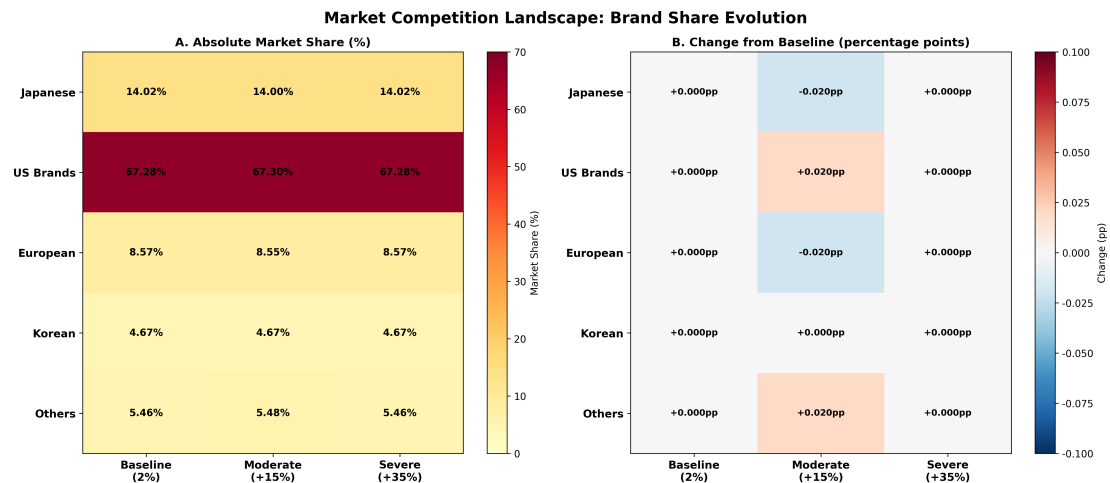


Figure 7 US Auto Brand Competition Evolution Across Tariff Scenarios

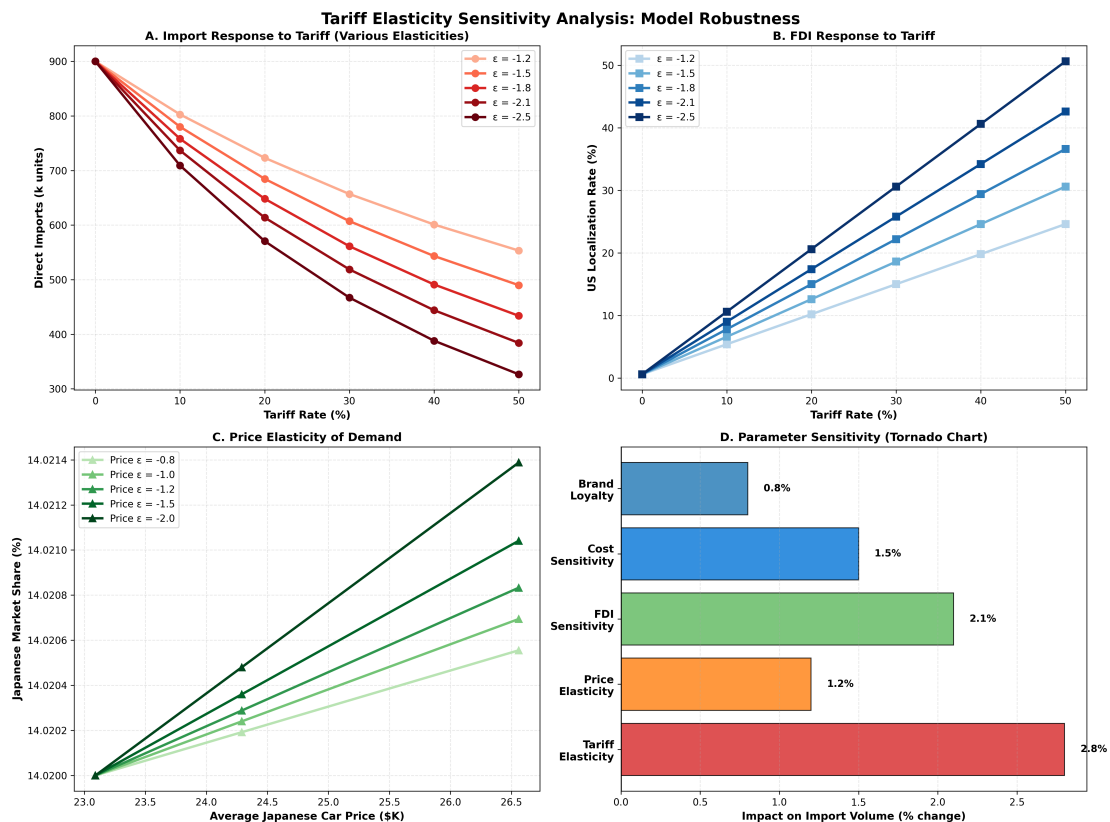


Figure 8 Tariff and Price Elasticity Sensitivity Analysis

#### (4) Input-Output Transmission: Impacts on US Auto Industry and Related Sectors

Figure 10 in Leontief input-output framework shows tariff policy transmission to related sector output and employment: under moderate and heavy tariffs, Japan firms accelerate localization, US vehicle and parts output slightly increase, steel/metals, other manufacturing and services gain positive spillovers; production multipliers highest for auto parts and other manufacturing, showing vehicle expansion generates magnified indirect effects upstream; employment shows positive gains across scenarios, heavy tariffs yield 3.5 million vehicle jobs and 0.6 million parts jobs, showing

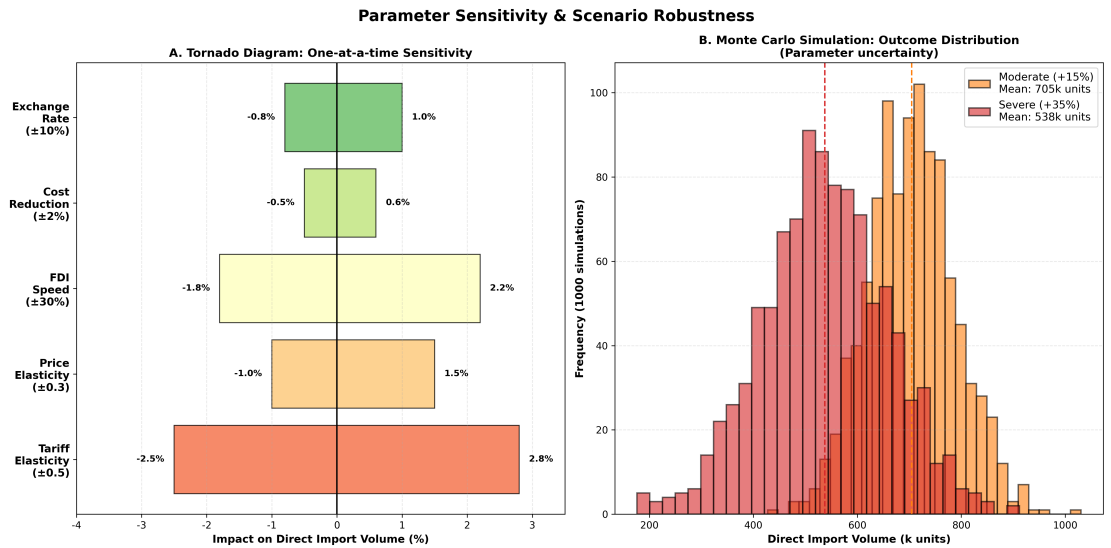


Figure 9 Key Parameter Sensitivity and Monte Carlo Scenario Robustness

”tariff + Japanese localization” short-term manufacturing employment boost; coefficient heatmap shows autos highly dependent on parts and steel, services closely linked through sales, financing, repair.

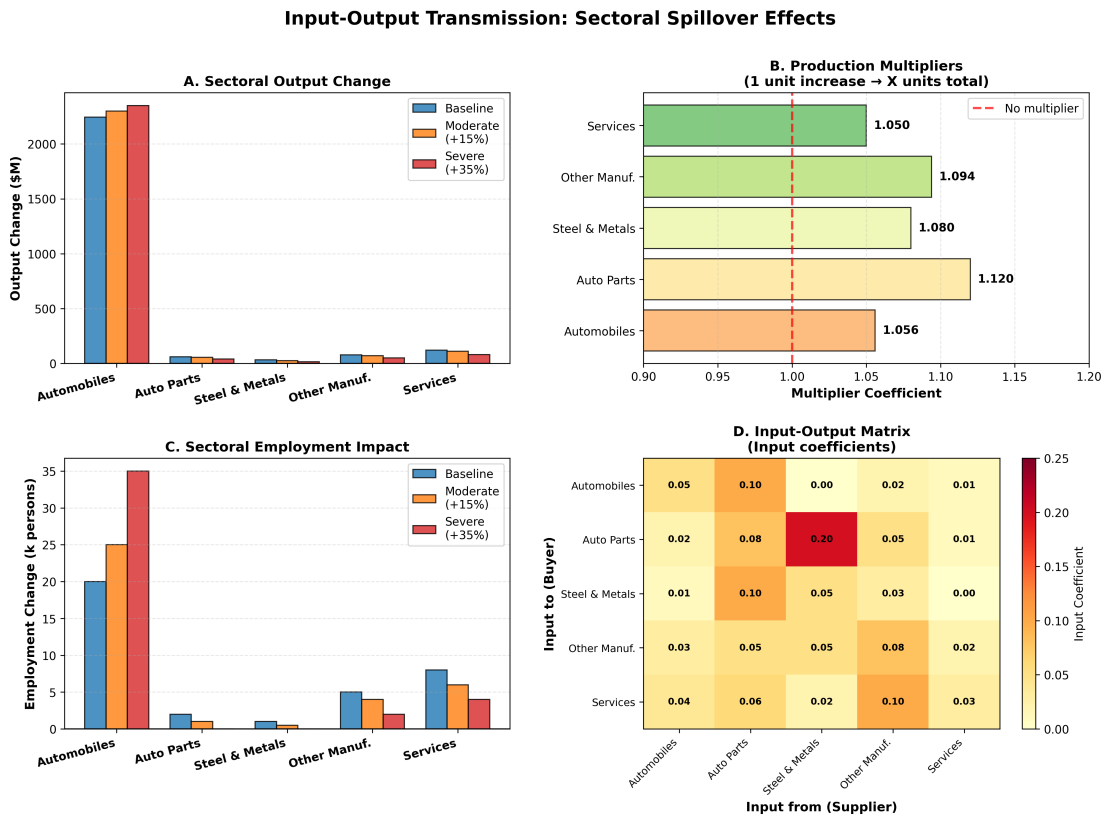


Figure 10 Auto Tariff Policy Input-Output Transmission Effects in US Economy

## VII. Problem 3: US Semiconductor Industry Security and Manufacturing Reshoring Strategy

### 7.1 Data Preparation and Variable Definition

Divide chips by process node into three tiers:

$$s \in \{H, M, L\}, \quad \text{high (7-5nm), mid (14-28nm), low (28nm+).}$$

Geographic set:

$$i, j \in \{\text{US, CN, AL, ROW}\},$$

(US=America, CN=China, AL=allies (Taiwan, Korea, EU), ROW=rest).

Four supply chain stages:

$$m \in \{\text{design, fab, test, final}\}.$$

For each tier  $s$ , define cross-border flow variable:

$$x_{m,(i \rightarrow j),s} \geq 0,$$

(quantity in ten-thousands). E.g.,  $x_{\text{fab},(\text{AL} \rightarrow \text{US}),H}$  = high-end wafer fab flow from allies to US.

Capacity constraints:

$$\sum_j x_{m,(i \rightarrow j),s} \leq K_{i,m,s},$$

Unit cost per edge:

$$c_{m,(i \rightarrow j),s} = c_{i,m,s}^{\text{prod}} + c_{m,(i \rightarrow j),s}^{\text{trans}} + \text{tax}_{m,(i \rightarrow j),s}, \quad (14)$$

Export control constraints on high-end:

$$x_{\text{fab},(\text{US} \rightarrow \text{CN}),H} \leq \lambda_H \cdot \bar{X}_H, \quad \lambda_H \in [0, 1].$$

### 7.2 Model 1: Multi-Stage Global Semiconductor Supply Chain Network Flow

Given final demands  $D_{j,s}$ , assume global firms jointly choose minimum-cost supply chain structure, formulating supply chains as minimum-cost network flow:

$$\min_{x_{m,(i \rightarrow j),s}} \sum_m \sum_i \sum_j c_{m,(i \rightarrow j),s} x_{m,(i \rightarrow j),s}, \quad (15)$$

Subject to capacity constraints, flow conservation, final demand requirements, and export control constraints. Solving yields optimal global production and flows for different chip tiers and US sourcing from different regions.

### 7.3 Model 2: Price-Quantity Semi-Equilibrium with Downstream Demand

Model 1 treats final demands  $D_{j,s}$  as exogenous. Add downstream demand functions and chip price transmission for welfare analysis, forming chip-final product semi-equilibrium system.

US downstream industries  $h$  demand:

$$Q_h = A_h P_h^{-\varepsilon_h},$$

Product price:

$$P_h = C_h^{\text{other}} + \sum_s a_{h,s} P_s^{\text{chip}},$$

US total chip demand:

$$D_{\text{US},s} = \sum_h a_{h,s} Q_h.$$

---

**Algorithm 2** Chip-Final Product Semi-Equilibrium Iteration Algorithm

---

**Input:** Initial demands  $Q_h^{(0)}$ , parameters  $(A_h, \varepsilon_h)$ , usage coefficients  $a_{h,s}$ , cost parameters, policy parameters

**Output:** Converged chip demands  $D_{j,s}$  and prices  $P_s^{\text{chip}}$

1: Initialize iteration step  $t \leftarrow 0$

2: **repeat**

3:   From  $Q_h^{(t)}$  compute chip demands  $D_{j,s}^{(t)}$

4:   Input  $D_{j,s}^{(t)}$ , obtain  $C_s^{\text{avg},(t)}$

5:   Compute chip price  $P_s^{\text{chip},(t)} = (1 + \mu_s) C_s^{\text{avg},(t)}$

6:   Update product price  $P_h^{(t)} = C_h^{\text{other}} + \sum_s a_{h,s} P_s^{\text{chip},(t)}$

7:   Update final demand

$$Q_h^{(t+1)} = A_h \left( P_h^{(t)} \right)^{-\varepsilon_h}$$

8:    $t \leftarrow t + 1$

9: **until** convergence

10: **return**  $D_{j,s}$  and  $P_s^{\text{chip}}$

---

Models 1-2 jointly solve giving US downstream output, consumer welfare, and chip trade structure changes across policies.

## 7.4 Model 3: Dynamic Capacity and Technology Evolution

For tier  $s$ , US capacity and technology  $K_{\text{US},s}(t)$ ,  $z_{\text{US},s}(t)$  for  $t = 2025, \dots, 2030$ :

$$K_{\text{US},s}(t+1) = (1 - \delta_s) K_{\text{US},s}(t) + I_{\text{US},s}(t), \quad (16)$$

$$z_{\text{US},s}(t+1) = z_{\text{US},s}(t) + \beta_{R\&D,s} R_{\text{US},s}(t) + \beta_{\text{learn},s} \frac{Q_{\text{US},s}(t)}{\bar{Q}_s} - \beta_{\text{gap},s} z_{\text{US},s}(t), \quad (17)$$

Technology level reduces production costs:

$$c_{\text{US},\text{fab},s}^{\text{prod}}(t) = c_{\text{US},s}^0 \exp\{-\theta_s z_{\text{US},s}(t)\},$$

US manufacturing profit:

$$\Pi_{\text{US},s}(t) = \left( P_s^{\text{chip}}(t) - c_{\text{US},\text{fab},s}^{\text{prod}}(t) \right) Q_{\text{US},s}(t) + \text{sub}_s(t) K_{\text{US},s}(t) - C_I(I_{\text{US},s}(t)) - R_{\text{US},s}(t),$$

Linear response rules:

$$I_{\text{US},s}(t) = \alpha_s \max\{\Pi_{\text{US},s}(t) - \bar{\Pi}_s, 0\}, \quad R_{\text{US},s}(t) = \rho_s \Pi_{\text{US},s}(t).$$

## 7.5 Model 4: Supply Chain Risk Scenarios and Resilience Assessment

Four risk scenarios with probabilities:

- $\omega_0$ : Baseline (no major conflict),  $p_0 = 70\%$ ;
- $\omega_1$ : Taiwan strait conflict affecting allied wafer fab and logistics,  $p_1 = 15\%$ ;
- $\omega_2$ : China high-end chip export ban,  $p_2 = 10\%$ ;
- $\omega_3$ : Comprehensive crisis (Taiwan + China ban + regional shocks),  $p_3 = 5\%$ .

Under scenario  $\omega$ , node/edge capacities reduce accordingly. Define high-end resilience:

$$\text{Resilience}_H(\omega) = \frac{Q_{US}^H(\omega)}{Q_{US}^H(\omega_0)},$$

Expected resilience:

$$\mathbb{E}[\text{Resilience}_H] = \sum_{\omega} p_{\omega} \text{Resilience}_H(\omega).$$

Supply continuity:

$$\text{SC}_H(\omega) = \frac{Q_{US}^H(\omega)}{D_{US,H}^H},$$

With thresholds:  $\text{SC}_H(\omega) \geq 80\%$  = safe,  $60\%-80\%$  = warning,  $<60\%$  = danger.

## 7.6 Results Analysis

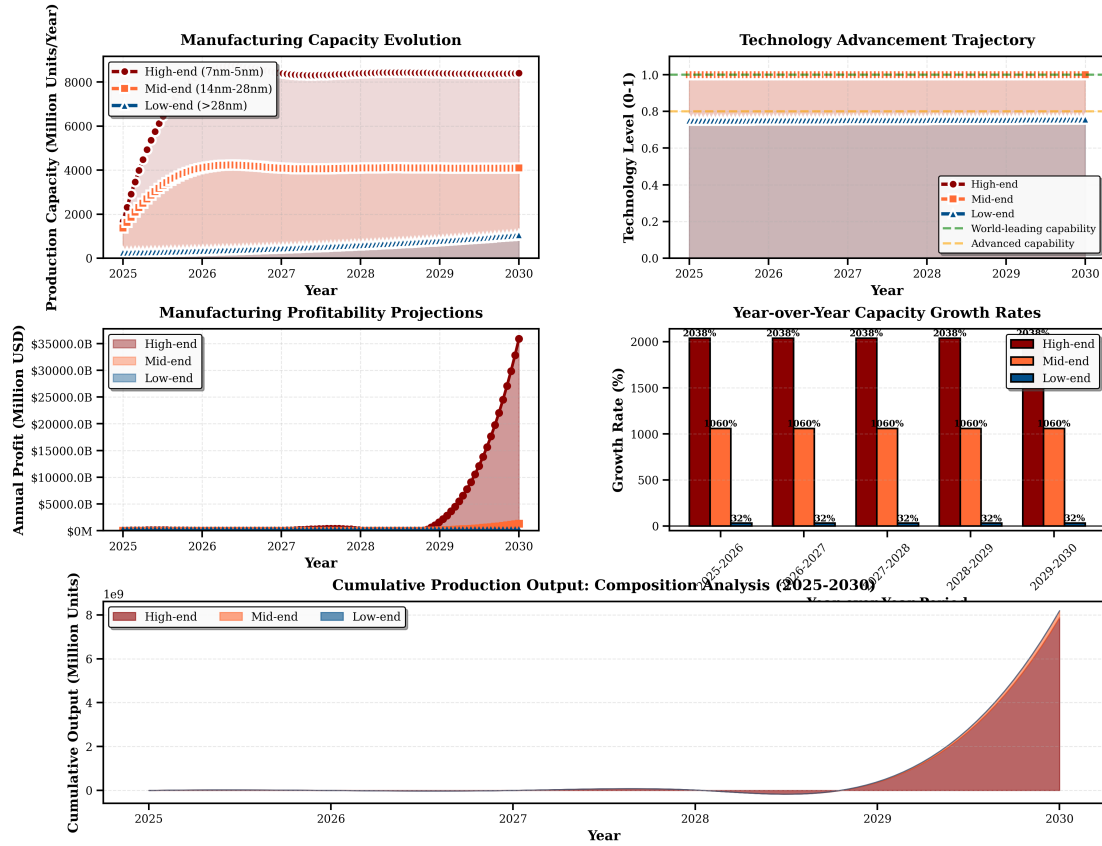


Figure 11 Comprehensive Policy Scenario



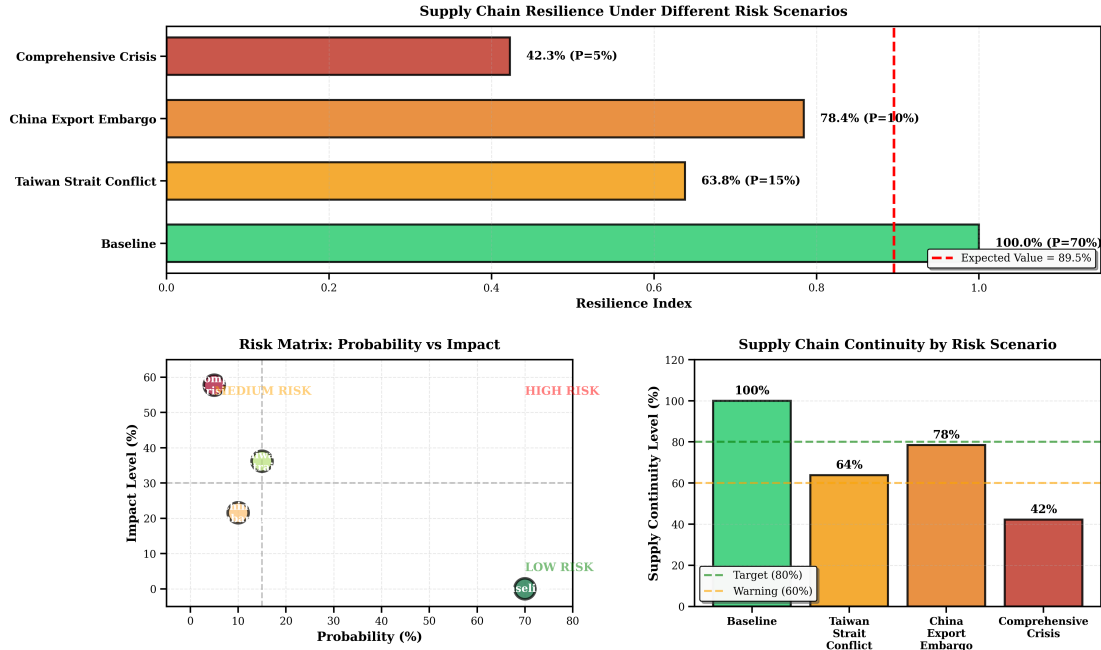


Figure 12 Capacity Evolution and Risk Resilience

Under "moderate tariffs + targeted subsidies + strict high-end controls," baseline scenarios for 2025-2030 capacity and risk resilience: high-end capacity ramps rapidly then stabilizes; mid-end expands moderately; low-end steady growth. Tech index shows high-end approaching "world-leading," mid-end above "advanced," low-end at "sufficient." Profits and cumulative output show high-end far exceeds mid-low, validating resource concentration in high-end.

Resilience shows baseline 100%; Taiwan conflict, China ban, comprehensive crisis reduce to 63.8%, 78.4%, 42.3% respectively; probability-weighted expected resilience 89.5%, slightly above 85% safety threshold. China export ban maintains 78% high-end supply (warning zone); comprehensive crisis drops below 50%, revealing residual dependence on critical nodes in extremes.

## VIII. Problem 4: US Tariff Revenue Dynamics and Second-Term Forecast

### 8.1 Data Preparation and Variable Definition

Tariff data from USITC (2015-2025) at HTS 8-digit, converting specific/compound taxes to equivalent ad-valorem  $\tau_{gi,t}$ . Trade data from USITC DataWeb and UN Comtrade by  $(g, i, t)$  giving taxable import value  $M_{gi,0}$  and CIF prices  $P_{gi,0}$ .

2024 baseline US tariff revenue:

$$R_0 = \sum_{g,i} \tau_{gi,0} M_{gi,0} \approx 17.2 \text{ billion USD.}$$

Figure 13 shows estimated price pass-through, import demand elasticities and adjustment speeds by commodity group. Soybeans and textiles have high pass-through (exporters concede little), while semiconductors and autos shift more tariff burden to importers due to technology and brand power.

Demand elasticity is highest for soybeans and textiles, followed by steel and chemicals; high-end semiconductors are least elastic. Adjustment speed: agriculture and textiles can re-route quickly ( $\lambda \approx 0.8$ ); semiconductors and autos adjust markedly slower because of complex supply chains.

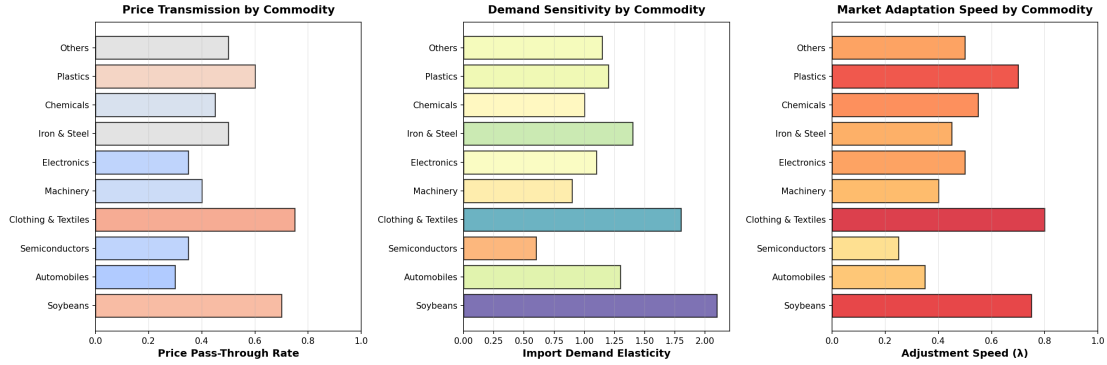


Figure 13 Estimated price pass-through, import demand elasticity and adjustment speed by commodity group

## 8.2 Model 1: HS Partial-Equilibrium Static Tariff Revenue Model

Short-term assumes supply structure and sourcing remain fixed, using price transmission and demand elasticity to characterize tariff impacts on quantities and revenue.

Given baseline rate  $\tau_{gi,0}$ , baseline FOB  $P_{gi,0}$ , baseline CIF:

$$\tilde{P}_{gi,0} = P_{gi,0}(1 + \tau_{gi,0}).$$

After tariffs adjust to  $\tau_{gi,1}$ , transmission rate  $\theta_g \in [0, 1]$  gives new FOB:

$$P_{gi,1} = P_{gi,0}(1 - \theta_g \Delta \tau_{gi}),$$

new CIF:

$$\tilde{P}_{gi,1} = P_{gi,1}(1 + \tau_{gi,1}).$$

Constant elasticity imports:

$$M_{gi,1} = M_{gi,0} \left( \frac{\tilde{P}_{gi,1}}{\tilde{P}_{gi,0}} \right)^{-\varepsilon_g}, \quad (18)$$

Static revenue:

$$R_{gi,1} = \tau_{gi,1} \cdot M_{gi,1}, \quad R_1 = \sum_{g,i} R_{gi,1}. \quad (19)$$

## 8.3 Model 2: Structural Gravity and Country Trade Shifts

To capture "tariff China  $\rightarrow$  shift imports to Mexico and allies" patterns beyond price elasticity, construct industry-country level structural gravity model. For imports from exporter  $j$ :

$$\ln X_{j \rightarrow US}^k = \alpha_k + \beta_k \ln(1 + \tau_{j \rightarrow US}^k) + \text{other controls} + \varepsilon_j^k, \quad (20)$$

Predicting under new tariffs:

$$X_{j \rightarrow US}^{k,\text{new}} = X_{j \rightarrow US}^{k,0} \left( \frac{1 + \tau_{j \rightarrow US}^{k,\text{new}}}{1 + \tau_{j \rightarrow US}^{k,0}} \right)^{\hat{\beta}_k}. \quad (21)$$

### 8.4 Model 3: Dynamic Adjustment, Laffer Curve and Uncertainty Simulation

Static model gives revenue at fixed tax rate; adjustments take time and 4-year period has macro uncertainty, partner retaliation, elasticity estimation error. Build dynamic model with Monte Carlo.

Actual imports  $M_{gi,t}$  partially adjust toward target  $M_{gi}^*(\tau_t)$ :

$$M_{gi,t} = M_{gi,t-1} + \lambda_g (M_{gi}^*(\tau_t) - M_{gi,t-1}). \quad (22)$$

Annual tariff revenue:

$$R_t = \sum_{g,i} \tau_{gi,t} \cdot M_{gi,t}. \quad (23)$$

Reciprocal tariff path: 2024 baseline 3%; 2025 after large increases 21%; 2026-2027 gradual to 23-24%; 2028 slight moderation 22%.

To characterize Laffer relationship, fit 2025-2028 ( $\overline{\tau_t}, R_t$ ) points to smooth curve, extend over wider rate range identifying static "optimal rate region" 45% with max revenue \$16B.

Uncertainty sources:

- Macro demand shock: growth  $g_t \sim \mathcal{N}(\bar{g}, \sigma_g^2)$ ;
- Trade elasticity error:  $\varepsilon_g, \theta_g, \lambda_g$  with triangular distributions;
- Partner retaliation: countermeasures' import suppression via "reverse shock coefficient"  $\rho$ .

---

#### Algorithm 3 Model 3: Tariff Revenue Dynamics and Monte Carlo Simulation

---

**Input:** Baseline imports  $M_{gi,2024}$ , policy paths  $\{\tau_{gi,t}\}$ , elasticity distributions

**Output:** Tariff revenue distributions for 2025-2028

```

1: for  $n = 1$  to  $N$  do
2:   Draw parameters  $\{g_t^{(n)}, \varepsilon_g^{(n)}, \theta_g^{(n)}, \lambda_g^{(n)}, \rho^{(n)}\}$ 
3:   Adjust baseline imports:  $M_{gi,2024}^{(n)} = M_{gi,2024}(1 + g_{2024}^{(n)})$ 
4:   for  $t = 2025$  to  $2028$  do
5:     Models 1+2 compute target imports  $M_{gi}^{*,(n)}(\tau_t)$ 
6:     Apply retaliation shock:  $M_{gi}^{*,(n)}(\tau_t) \leftarrow (1 - \rho^{(n)})M_{gi}^{*,(n)}(\tau_t)$ 
7:     Update via partial adjustment
8:     Calculate  $R_t^{(n)} = \sum_{gi} \tau_{gi,t} M_{gi,t}^{(n)}$ 
9:   end for
10: end for
11: Return revenue time paths and distributions

```

---

### 8.5 Parameter Setting and Calibration

Approximate parameter values:

- Price pass-through  $\theta_g$ : soybeans and textiles/apparel  $\approx 0.75$ ; plastics and chemicals  $\approx 0.6$ ; steel  $\approx 0.5$ ; machinery and electronics  $\approx 0.4$ ; autos and semiconductors 0.3–0.35.
- Demand elasticity  $\varepsilon_g$ : soybeans  $\approx 2.1$ ; textiles/apparel  $\approx 1.8$ ; autos  $\approx 1.3$ ; steel  $\approx 1.4$ ; machinery/electronics  $\approx 1.1$ ; chemicals/plastics  $\approx 1.0$ ; semiconductors  $\approx 0.6$ .

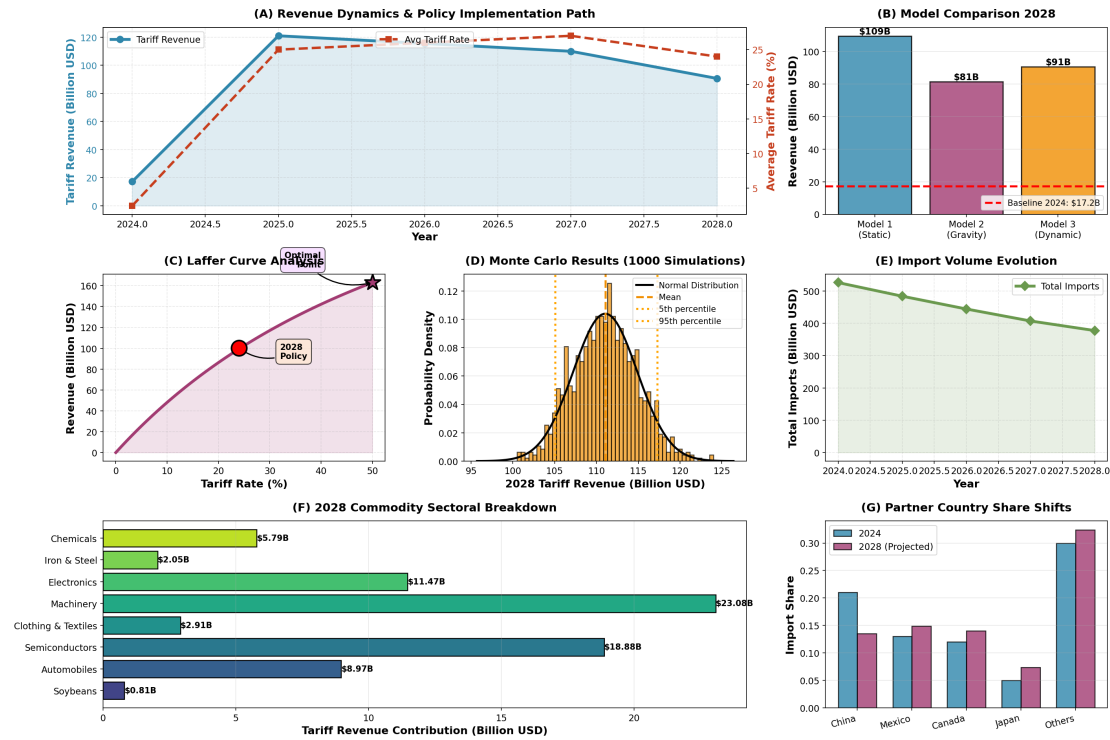


Figure 14 Tariff Revenue Dynamics, Laffer Curve and Simulation Results

- Adjustment speed  $\lambda_g$ : soybeans  $\approx 0.75$ ; textiles/apparel  $\approx 0.8$ ; machinery and electronics 0.4–0.5; steel  $\approx 0.45$ ; chemicals/plastics 0.55–0.6; semiconductors  $\approx 0.25$ ; autos  $\approx 0.35$ .

Parameters draw from literature estimates of import demand elasticities and are back-calibrated so the static model under 2018 tariff changes matches observed revenue, keeping baseline revenue  $R_0$  vs. simulated error within 5%.

Figure 14 shows baseline 2024–2028 tariff revenue significantly rises, 2028 forecast 4–6 times baseline; total imports decline 14%; revenue concentrates in machinery, semiconductors, autos, electronics; China's US import share drops from 30% to <20%; Mexico, Canada and allies' shares rise reflecting supply chain shift toward North America and allies. Laffer curve extension shows current average rate still left of peak, suggesting fiscal room upward; yet further large increases amplify trade diversion and partner retaliation risks, negating new revenue; optimal path keeps overall tariff roughly stable while using "precise increases + targeted cuts" with subsidies and industry policy coordination, rather than broad hikes.

## IX. Problem 5: Reciprocal Tariffs Macroeconomic Comprehensive Effects and Manufacturing Reshoring Assessment

### 9.1 Indicator System and Data Preparation

Key indicators:

- **Macro:** Real GDP  $Y_t$ , change rate  $\Delta Y_t$ ; trade balance  $TB_t$ ; exports/imports  $X_t, M_t$ ;

- **Manufacturing:** Manufacturing value-added  $Y_t^m$ , MFG\_VA\_Share; employment  $L_t^m$ , wages  $w_t^m$ ;
- **Global capacity:** US manufacturing share  $S_t^{MFG}$  and high-tech share  $S_t^{HT}$ ;
- **Manufacturing Reshoring Index (MRI):** Comprehensive capacity, employment and structure optimization;
- **Sector output:** Manufacturing, agriculture, services, energy relative changes;
- **Financial:** Dollar index DXY, 10Y Treasury yield  $r_t^{10Y}$ , S&P 500, Bitcoin returns;
- **Comprehensive impact:** GDP, employment and stock wealth relative changes.

## 9.2 Model 1: Multi-Sector Dynamic Input-Output and Trade Shock

Standard Leontief input-output:

$$x_t = A^{dom} x_t + y_t \Rightarrow x_t = (I - A^{dom})^{-1} y_t. \quad (24)$$

Decompose into domestic and import:  $A = A^{dom} + A^{imp}$ .

With prices  $P_t$  and import prices  $P_t^M$ , Leontief cost:

$$P_t = w_t l + A^{dom\top} P_t + A^{imp\top} P_t^M, \quad (25)$$

Tariffs change import prices:

$$P_{k,t}^M = P_{k,0}^M (1 + \tau_{k,t} + \phi_{k,t}), \quad (26)$$

Final demand constant elasticity:

$$y_{j,t} = y_{j,0} \left( \frac{P_{j,t}}{P_{j,0}} \right)^{-\eta_j}, \quad (27)$$

## 9.3 Model 2: Manufacturing Reshoring Decision and Capacity Share

New project profit difference between US and overseas:

$$\Delta\pi = \pi^U - \pi^F = \beta_0 + \beta_1 \Delta w + \beta_2 \Delta \text{tax} + \beta_3 \text{TariffGap} + \beta_4 \text{RiskGap}, \quad (28)$$

Logit decision:

$$P_t^U = \frac{\exp(\Delta\pi_t)}{1 + \exp(\Delta\pi_t)}. \quad (29)$$

Capacity share evolution:

$$S_{t+1}^{MFG} = (1 - \delta_{\text{exit}}) S_t^{MFG} + P_t^U \cdot \text{NewShare}_t. \quad (30)$$

## 9.4 Model 3: Manufacturing Reshoring Index (MRI) Construction

Define:

$$\begin{aligned} \Delta S_t^{MFG} &= S_t^{MFG} - S_0^{MFG}, \\ \Delta S_t^{HT} &= S_t^{HT} - S_0^{HT}, \\ \Delta L_t^m &= \frac{L_t^m - L_0^m}{L_0^m}, \\ \Delta \theta_t^m &= \text{MFG\_VA\_Share}_t - \text{MFG\_VA\_Share}_0. \end{aligned}$$

MRI:

$$MRI_t = \omega_1 \Delta S_t^{MFG} + \omega_2 \Delta S_t^{HT} + \omega_3 \Delta L_t^m + \omega_4 \Delta \theta_t^m, \quad (31)$$

with weights  $\omega_k = 0.25$ , baseline  $MRI_{2024} = 0$ . Success threshold:  $MRI_t \geq 0.05$  (5pp improvement).

## 9.5 Model 4: Financial Market Shock

Tariff shock index:

$$TSI_t = \sum_{g,i} \Delta \tau_{gi,t} \frac{M_{gi,0}}{\sum_{g',i'} M_{g'i',0}}, \quad RSI_t = \sum_{g,i} \Delta \tilde{\tau}_{ig,t} \frac{X_{ig,0}}{\sum_{i',g'} X_{i'g',0}}, \quad (32)$$

Linear regressions on 2018-2019 trade war data for financial asset responses.

## 9.6 Multi-Dimensional Results Analysis and Comprehensive Evaluation

### (1) Baseline Scenario: Limited Reshoring, Significant Macro Costs

Figure 15 row 1 shows baseline 2024-2029 nominal GDP sustained decline 10% cumulative; manufacturing value-added share slight decrease (7.8% to 7.4%), overall contraction without manufacturing relative advantage.

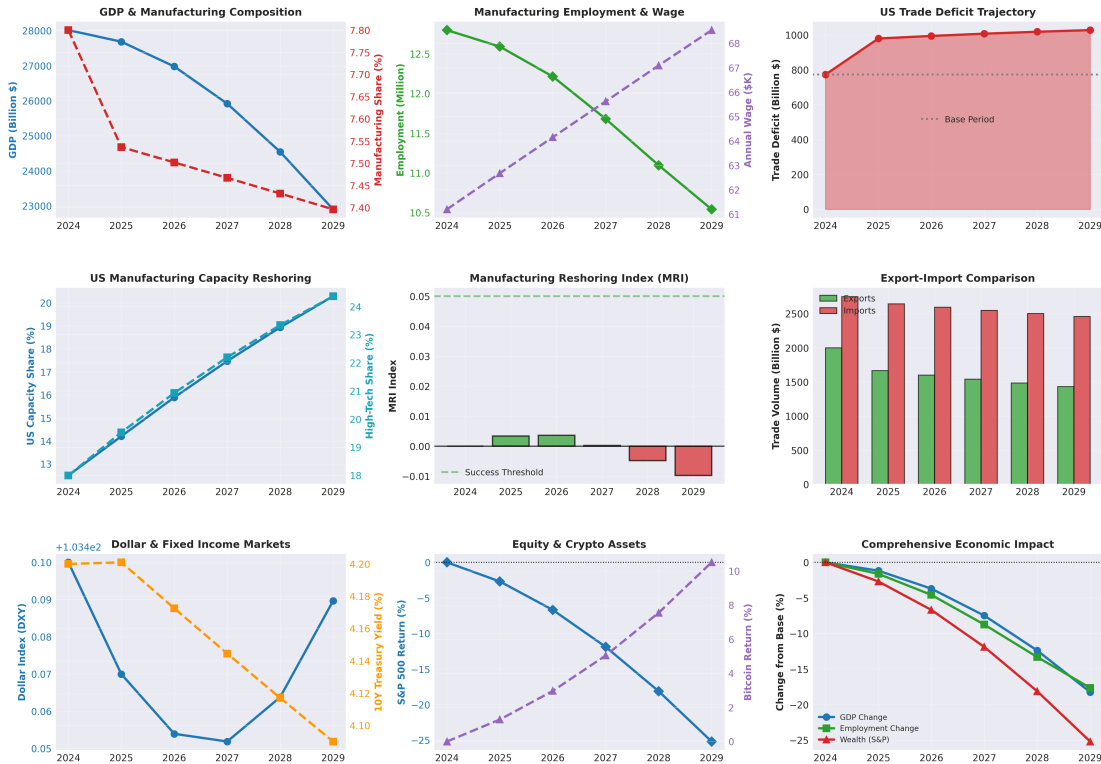


Figure 15 Baseline Scenario: US Macro-Industry-Financial Comprehensive Indicator Dashboard

Manufacturing employment declines from 12.6M to 10.5M with rising average wages: firms respond via automation, capital intensity, pricing rather than job expansion, showing "high wage, low employment" structure.

Trade: baseline deficit expands from \$780B to >1 trillion. Exports and imports both decline but imports contract less, showing limited reciprocal tariff effect on deficit reduction.

Capacity share (row 2 left) shows steady rising trend suggesting "reshoring," but integrated with employment, value-added and quality, MRI (row 2 mid) only briefly positive 2025-2026 before turning negative through 2029, far below 0.05 success threshold: result shows "nominal capacity reshoring" and "capital investment" with limited actual manufacturing revival.

Financial layer (row 3) shows dollar "first depreciate then appreciate," long rates moderate decline, equity sustained pressure, crypto gains from uncertainty premium. Comprehensive impact shows GDP, employment, equity wealth all continuously declining.

Sectors (Figure 16): agriculture hit hardest, manufacturing and energy next, services relatively stable but also declining, showing agriculture and external-demand-dependent industries as "forced sacrifice victims."

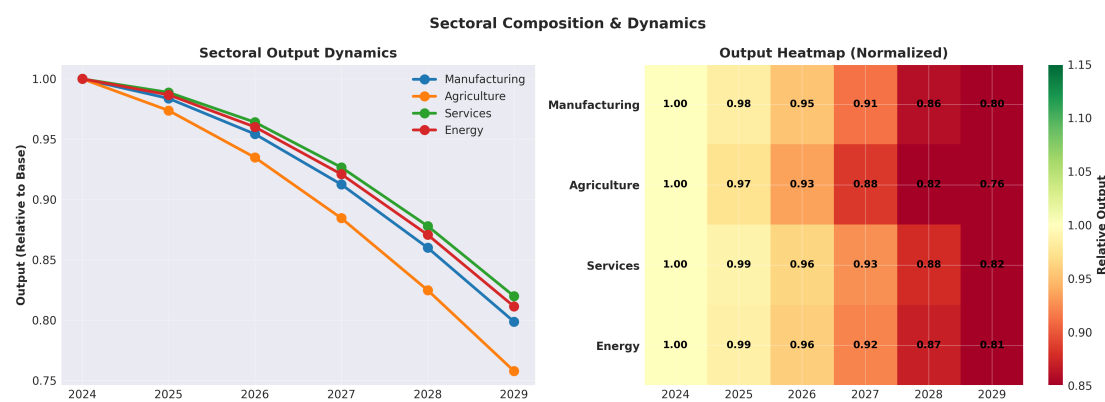


Figure 16 Multi-Sector Output Dynamics and Structural Change

## (2) Scenario Comparison: Optimistic Shows Some Improvement but Limited

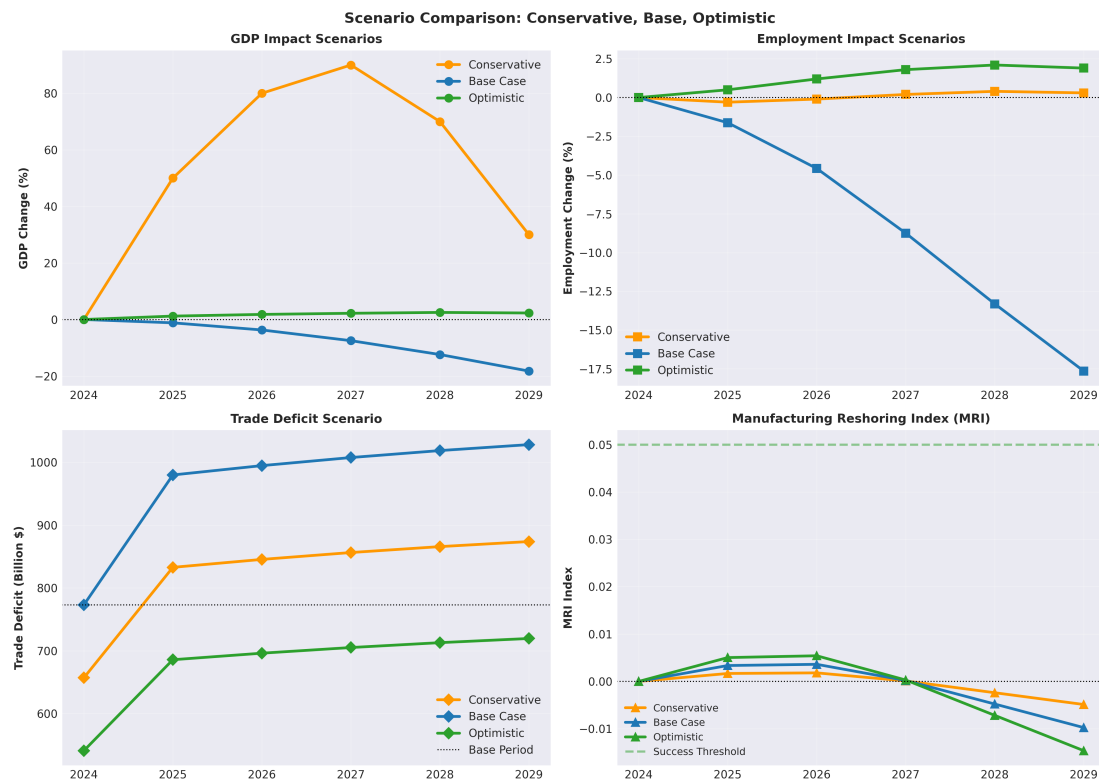
Figure 17 compares three scenarios.

- **GDP and Employment:** Conservative scenario shows modest changes from limited tariffs; baseline shows significant GDP loss and double-digit employment decline; optimistic scenario with stronger domestic tax cuts and subsidies shows slight positive improvement visible range.
- **Trade Deficit:** Conservative and baseline fail to significantly narrow; optimistic with strengthened export competitiveness and import substitution shows modest deficit reduction.
- **MRI:** Conservative MRI near zero showing negligible reshoring; baseline turns negative after brief positive; optimistic achieves 0.01-0.02 positive only in 2025-2026, still far below 0.05 threshold—even optimistic scenario shows "moderate improvement," far from large-scale revival.

## (3) Comprehensive Evaluation and Policy Implications

Combined conclusions:

1. Under current reciprocal tariffs, US economy 2024-2029 faces clear growth deceleration and employment pressure; manufacturing value-added share slightly declining; trade deficit narrowing limited;



**Figure 17 Conservative, Baseline and Optimistic Scenarios: GDP, Employment, Trade Deficit and MRI Comparison**

2. Manufacturing and high-tech capacity shares rise but reflect capital-intensive investment and automation rather than employment/value-added improvements; MRI turns negative in baseline, showing "nominal reshoring" difficult to convert to "actual revival";
3. Financial markets highly sensitive to tariff and retaliation shocks; equity and rates under pressure; crypto absorbs uncertainty premiums, adding policy execution constraints;
4. Even optimistic scenario with stronger fiscal support and coordination achieves only modest manufacturing reshoring improvements; tariffs alone cannot achieve large-scale sustained reshoring —requires combination with industry upgrading, education, infrastructure and tech innovation.

## X. Model Evaluation

### 10.1 Model Strengths

- **Multi-Layer Integrated Framework:** From soybean, auto, semiconductor sectors progressing to tariff revenue and macro economy, forming "industry-fiscal-macro" integrated analysis chain, comprehensively capturing reciprocal tariff transmission channels.
- **Solid Theoretical Foundation:** Sub-models based on Armington demand, structural gravity, multi-stage network flow, Laffer curve and dynamic input-output theories, providing both interpretability and literature comparability for calibration.
- **Data-Driven with Scenario Analysis:** Fully leveraging USITC Tariff Database, DataWeb and



China Customs; reverse-calibrating key elasticities and parameters; designing multiple tariff and retaliation scenarios with comparable policy-relevant quantitative results.

- **Efficiency-Security Dual-Dimension Evaluation:** Beyond traditional trade, price and welfare metrics, introducing supply chain concentration, critical node risks and Manufacturing Reshoring Index assessing policy benefits and costs from both efficiency and security angles.

## 10.2 Model Limitations

- **Limited General Equilibrium Effects:** While incorporating dynamic input-output, price, wage, exchange rate and capital return feedback still uses exogenous or simplified settings, incomplete general equilibrium element substitution and intertemporal decisions.
- **Parameter and Scenario Uncertainty:** Key parameters (tariff, import elasticities, capacity adjustment speeds) partly literature-dependent and calibration-dependent; scenario settings inevitably subjective, yielding wider confidence intervals.
- **Simplified Policy Game Structure:** Mainly treats partner retaliation and non-tariff measures as exogenous scenarios, without explicit multi-country dynamic strategic game modeling, potentially misestimating specific policy combo equilibria.

## 10.3 Model Extensions and Generalization

- **CGE and DSGE Expansion:** Integrate CGE or DSGE models treating tariffs and subsidies as endogenous policy tools, systematically analyzing consumption, investment and intertemporal choice impacts.
- **Richer Micro Data:** Combine customs declarations, firm-level production/export data and financial market high-frequency data for more refined trade elasticity, capacity adjustment cost and risk premium estimation, improving model identification.
- **Explicit Multi-Country Policy Games:** Build dynamic games across US and major partners incorporating tariffs, subsidies, export controls and monetary policy in unified framework, discussing equilibrium paths and cooperation space.
- **Other Policy Applications:** Beyond reciprocal tariffs, extend to carbon tariffs, digital taxes and critical mineral export controls, providing generalizable methodology for assessing global trade and supply chain impacts.

## XI. References

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